Personalized Fitness Recommendation System Based on Sensor Data

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***Abstract*—Human Activity Recognition (HAR) utilizing smartphone inertial sensors plays a vital role in the development of modern fitness tracking and healthcare solutions. However, the presence of variability in human movement patterns, sensor orientation, and personal patterns poses significant challenges for the development of effective activity classification solutions. In this context, the proposed research introduces a novel Hybrid RF-LSTM approach that utilizes the benefits of sequence modeling and ensemble-based classification. Long Short-Term Memory (LSTM) networks can effectively capture the sequence patterns and temporal dynamics that are intrinsic to inertial sensor-based activity classification. On the other hand, the Random Forest (RF) classifier can ensure the robustness of the proposed approach through ensemble learning and variance reduction. In the proposed approach, the UCI HAR dataset has been utilized to validate the proposed approach. From the results, the proposed approach can ensure effective multiclass classification with robust generalization. Compared to the proposed approach, the use of attention-based deep learning approaches can ensure a trade-off between complexity and accuracy, making the proposed hybrid approach more viable for the development of effective activity classification solutions.**

***Index Terms*—Human Activity Recognition, HAR , Hybrid RF-LSTM, Long Short-Term Memory, Random Forest, Ensemble Learning, Inertial Sensor Data, UCI HAR Dataset**

I. INTRODUCTION

Human Activity Recognition (HAR) has been identified as a key area of research in fitness monitoring and healthcare applications, allowing for the automatic recognition of human activities using wearable and smartphone sensor data [1], [2]. Modern smartphones have been observed to carry several inertial sensors, including accelerometers and gyroscopes, which collect multivariate data continuously [1]. Despite significant advances in the field of HAR, the presence of differences in human activities, sensor placement, and motion intensity continues to affect robustness and generalization performance [3]. As a result of the sequential nature of human activities, an efficient HAR system should focus on temporal dependencies while maintaining computational efficiency. Previous HAR systems employed a manual observation and rule-based approach, which was time-consuming and difficult to scale up to larger systems [3], [4]. The emergence of smartphones and wearable devices has resulted in the increased use of machine learning-based HAR systems in fitness monitoring and healthcare applications [2]. The use of traditional machine learning techniques, such as Random Forest, is observed to have strong robustness and interpretability in sensor data-based classification tasks, but their performance is limited to a certain level of temporal modeling capability [6], [7]. The use of deep learning techniques, such as LSTM, is observed to have strong temporal modeling capabilities, but their performance is limited to a certain level of variance [8].In order to overcome the limitations of traditional machine learning techniques, a novel Hybrid RF-LSTM framework is proposed, which combines temporal sequence modeling with decision-level machine learning techniques. The proposed framework is expected to result in efficient HAR systems while maintaining robustness in multiclass activity recognition tasks while keeping computational complexity moderate to allow for real-time applications in healthcare monitoring systems.

II. LITERATURE REVIEW

Human Activity Recognition (HAR) has garnered significant research interest due to its applications in fitness tracking, healthcare monitoring, assisted living, and smart environments [2], [3]. Initially, HAR was performed using conventional machine learning algorithms like k-Nearest Neighbors, Sup-

port Vector Machines, Decision Trees, and Random Forests using hand-engineered time and frequency domain features from inertial sensor data [1], [5], [6], [8]. Such techniques provided reliable classification performance with reasonable computational costs and interpretability, making them suitable for deployment on mobile and handheld devices [6] . As computational power increased, deep learning techniques were employed. Convolutional Neural Networks (CNNs) were employed for the extraction of relevant features from accelerometer and gyroscope sensor data, while Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, were also shown to be effective for the classification of sequential sensor data [9], [10]. Combination of CNNs and LSTMs was also shown to improve classification accuracy on standard HAR datasets like UCI-HAR and WISDM. However, deep learning architectures require large amounts of training data and incur significant computational costs. Additionally, they lack interpretability owing to the black-box nature of deep learning models [9], [10] Recently, the use of multimodal and sensor fusion techniques has been employed to improve the robustness of the HAR systems. However, this comes at the cost of increased complexity and power requirements. Though deep learning architectures possess significant representational power, classical Random Forests were also shown to be effective for the task of HAR due to its robustness to noise and its ability to provide feature importance interpretability [6], [7]. Motivated by the recent advances in sequential deep learning and ensemble-based classification algorithms, a novel framework called Hybrid RF-LSTM is proposed. The LSTM network is employed for the extraction of temporal motion dynamics present in the inertial sensor data, while the Random Forest classifier is employed for the classification task to improve the overall performance through ensemble averaging and variance reduction.

III. BACKGROUND

It describes the methodology for activity recognition, the technique used, and the collection and preprocessing steps that were performed [3].

A. Activity Recognition Method

1) Multi-Class Activity Classification for Human Activity Recognition: The model classifies smartphone inertial data into multiple activity classes, using segmented windows assigned to one of the following classes [5]. Each segmented sensor window is assigned to one of the following categories:

(a) Walking
(b) Walking Upstairs
(c) Walking Downstairs
(d) Sitting
(e) Standing
(f) Laying

This formulation allows the recognition of both dynamic and static activities, thus enabling extensive monitoring of motion patterns [2].

2) Operation: The operational flow of the activity recognition system is summarized as follows:

(a) **Signal Acquisition:** Tri-axial accelerometer and gyroscope signals are captured using a waist-mounted smartphone at a sampling frequency of 50 Hz [5].
(b) **Window Segmentation:** The raw sensor signals are segmented into fixed-length windows of 2.56 seconds with 50% overlap to preserve temporal continuity and transitional activity patterns [5].
(c) **Feature Representation:** Standard time- and frequency-domain features are extracted from segmented sensor windows, as detailed in Section IV.

3) Output: The trained classifier assigns probability scores to all activity classes and predicts the activity with the highest likelihood [6].

4) Benefits: The proposed activity recognition approach offers the following advantages:

(a) **Accurate Recognition:** Effectively distinguishes between static and dynamic activities using motion patterns [6].
(b) **Real-Time Applicability:** Enables low-latency activity prediction suitable for mobile and edge devices [1].
(c) **Personalized Monitoring:** Supports activity-based fitness analysis and personalized recommendation generation [2].

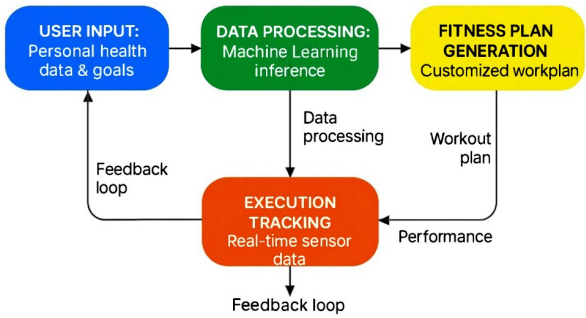

Fig. 1: Proposed System Architecture showing complete workflow from User Input to Feedback Loop

B. System Architecture

Fig. 1 illustrates a four-module closed-loop framework [6]. The user input module collects personal health data, historical activity logs, and fitness goals to establish baseline context for personalized inference. The data processing module integrates a central Hybrid RF-LSTM based activity recognition engine that processes real-time sensor signals and user inputs through noise removal, normalization, and feature extraction before model application [7].The inference block identifies performed activities and generates analytical insights for recommendation construction. The fitness plan generator uses inference outputs to create customized workout plans specifying activities, intensity, and duration aligned with user goals. The execution tracking module monitors accelerometer and gyroscope data during activity to evaluate adherence, detect deviations, and assess movement quality [4].Two internal

feedback loops enhance adaptability: (i) execution tracking to data processing to update the inference model with new data, and (ii) execution tracking to fitness plan generator to refine future recommendations.Overall, the architecture enables continuous learning, personalized adaptation, and real-time guidance through interconnected modules and adaptive feedback mechanisms.

C. Methodology and Data Collection

1) Dataset Overview: The approach utilizes the UCI HAR dataset, a benchmark dataset for smartphone-based Human Activity Recognition (HAR) [5]. Data were collected from 30 volunteers (ages 19–48) performing six activities: walking, walking upstairs, walking downstairs, sitting, standing, and lying. Signals were recorded using a Samsung Galaxy S II smartphone worn at the waist, capturing three-axis linear acceleration and angular velocity at 50 Hz.Preprocessing follows the standard UCI HAR pipeline: signals are segmented into 2.56-second windows with 50% overlap, labeled by activity, and normalized [5]. A total of 561 time- and frequency-domain features were extracted [8]. The dataset was split into 80% training and 20% testing subsets with balanced class distribution. Table I summarizes the main characteristics of the dataset.

TABLE I: Summary of UCI Human Activity Recognition Dataset

Attribute	Description
Number of Subjects	30
Activities	Walking, Upstairs, Downstairs, Sitting, Standing, Laying
Sensors Used	Accelerometer, Gyroscope
Sampling Rate	50 Hz
Window Size	2.56 s (128 readings)
Total Features	561
Train/Test Split	80% / 20%
Data Format	CSV (processed and labeled)

2) Data Preprocessing Steps: The following preprocessing steps are applied prior to model training:

- **Noise Filtering:** A low-pass Butterworth filter separates gravity and body-acceleration components [3].
- **Normalization:** Sensor signals are normalized to ensure consistent scaling across axes [8].
- **Assignment of Labels:** Each window is labeled with the corresponding activity class [5].

3) Feature Extraction: Time-domain features-mean, standard deviation, minimum, maximum, median absolute deviation-and frequency-domain features-FFT-based descriptors-are extracted, resulting in a total of 561 features per instance [8].

4) Dataset Partitioning: The dataset is divided into two subsets for experimentation:

- **Training Set (80%):** Used for training the Random Forest classifier [6].
- **Test Set (20%):** Used for evaluating model performance [5].

This partitioning ensures balanced representation of all activity classes across training and testing phases.

5) Sensor Data Characteristics: Smartphone inertial sensor data possesses comprehensive motion information that forms a significant basis for human activity recognition [3]. The combination of accelerometer and gyroscope signals ensures capture of linear and angular movements, facilitating accurate activity recognition in fitness monitoring and behavioral analysis [2].

IV. PROPOSED FRAMEWORK AND MODEL

This section gives an overview of the proposed Hybrid RF-LSTM model framework for Human Activity Recognition (HAR), starting with data collection to personalized recommendations [6]

• Sensor Signal Acquisition: Records the patterns of motion using tri-axial accelerometer and gyroscope data from smartphone sensors [1], [5].

• Preprocessing: noise filtering, normalization, and segmentation enhance the signal quality and consistency [3], [8].

• Feature Extraction: Standard time- and frequency-domain features are extracted from segmented sensor windows, as detailed in Section IV [8].

• Feature Selection: Uses Random Forest feature-importance measures to reduce the dimensionality while retaining the most discriminative features [7].

• Classification Layer: This layer performs multiclass activity classification using the Hybrid RF-LSTM framework. [6].

This framework achieves an effective activity recognition with low computational overhead and high interpretability [6], [7]. A detailed feature representation and processing steps can be obtained from Table I.

A. Proposed Model

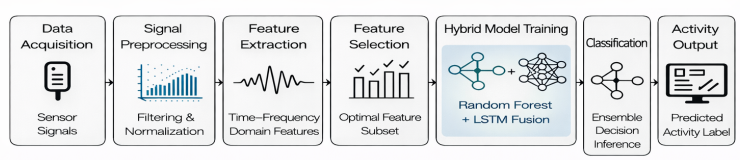

Fig. 2: Proposed Model Diagram for applying Hybrid RF-LSTM on UCI HAR Dataset showing the complete workflow from sensor data acquisition to personalized recommendations

Fig. **??** presents the proposed Human Activity Recognition (HAR) model diagram, displaying the complete workflow of sensor data acquisition through to personalized recommendations. Emphasis is placed on learning discriminative boundaries from engineered features, rather than raw signals themselves, with ensemble learning exploited for robustness and interpretability [6], [7].

The detailed explanation of the proposed Human Activity Recognition (HAR) model illustrated in Fig.**??** is provided as follows.

B. Data Collection

Sensor data are acquired from smartphones equipped with tri-axial accelerometers and gyroscopes, which continuously capture motion signals representing user activities and serve as raw system input [1].

C. Preprocessing

Collected signals undergo noise reduction and normalization to remove sensor distortions and axis-scale variations [3], [8]. This ensures consistent, high-quality inputs for downstream processing.

D. Feature Extraction

Standard time- and frequency-domain features are extracted from segmented sensor windows, as described in Section IV [8].

E. Feature Selection

To reduce dimensionality and redundancy, Random Forest feature-importance scores are applied [7]. Only the most informative features are retained, improving computational efficiency and generalization performance.

F. Feature Selection Analysis and Ablation Study

The UCI HAR dataset contains 561 handcrafted features, including statistical measures (mean, standard deviation, signal magnitude area, entropy, correlation) and frequency-domain spectral energy derived via FFT. High dimensionality increases computational cost and redundancy.

An ablation study compared model performance with and without feature selection. Features were ranked using Random Forest impurity-based importance, and the top 150 features were selected for training, while the baseline model used all 561 features

TABLE II: Ablation Study on Feature Selection

Model Setting	Features	Accuracy (%)	Cost
Without Feature Selection	561	97.42	High
With Feature Selection (Top 150)	150	98.78	Low

Table II presents the comparison results. The feature-selected model achieved higher accuracy while reducing computational overhead, confirming that feature selection improves both efficiency and generalization. This validates the effectiveness of the proposed feature selection strategy and avoids unnecessary model complexity.

G. Model Training

The feature set and structured sequences of sensors are used to train the suggested model of Hybrid RF and LSTM. First, a supervised learning model of a Long Short-Term Memory (LSTM) is used to train a network to learn the temporal dependencies and dynamic patterns of motion using sequential information from the inertial sensors [8]. This model learns the high-level representations of the temporal patterns, which capture the variability of the activity transitions and motion patterns.The representations of the temporal patterns are then used to train a Random Forest (RF) classifier [6]. This is a model of ensemble learning that combines a variety of decision trees, which are trained using different feature sets. This model is used to obtain a stable model of decision through the use of bagging and ensemble methods.

H. Classification

In the inference phase, the incoming sequence of sensor values is passed through the trained LSTM network to obtain the temporal feature embeddings. The feature embeddings are then passed through the Random Forest classifier for the final prediction of activities. The class label is obtained through a majority vote among the individual decision trees in the Random Forest, thereby providing a robust classification performance [6].

I. Activity Output

The model outputs the predicted activity label corresponding to the performed action. This output supports real-time activity monitoring, visualization, and personalized fitness recommendations [2].

J. Mathematical Preliminaries

This subsection presents the mathematical formulations used during preprocessing, feature extraction, and model evaluation.

1) **Signal Transformation:** The preprocessing stage involves applying a transformation function to the raw sensor signal to remove noise and separate relevant components. The transformed signal S_{proc} is expressed as

$$S_{\text{proc}} = T(\alpha, \beta, \gamma) \cdot S \tag{1}$$

where S represents the original sensor signal, and $T(\alpha, \beta, \gamma)$ denotes the transformation function incorporating filtering, scaling, and normalization parameters.

2) **Frequency-Domain Representation:** Frequency-domain features are extracted using the Fast Fourier Transform (FFT). The discrete Fourier transform of a signal $x(n)$ is given by

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \tag{2}$$

where N denotes the window length and k represents the frequency index.

3) **Feature Importance Estimation:** The importance of each feature in the Random Forest classifier is computed based on the average decrease in impurity across all decision trees, expressed as

$$FI(f_i) = \frac{1}{T} \sum_{t=1}^{T} \Delta I_t(f_i) \tag{3}$$

where $FI(f_i)$ denotes the importance of feature f_i, T is the total number of trees, and $\Delta I_t(f_i)$ is the impurity reduction contributed by feature f_i in tree t [6], [7].

4) **Classification Accuracy:** The performance of the proposed model is evaluated using classification accuracy, defined as

$$\text{Accuracy} = \frac{\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i)}{N} \tag{4}$$

where y_i and $\hat{y}_i$ denote the true and predicted class labels, respectively, N is the total number of samples, and $\mathbb{I}(\cdot)$ is the indicator function.

V. RESULTS AND DISCUSSION

This section accommodates the model performance summary, metrics evaluation, visual trends, accuracy plots, loss plots, and observed issues related to the proposed Human Activity Recognition (HAR) system [5].

A. Model Performance Summary

• Training Accuracy: Started at around 70.12% and grew to 98.35%, reflecting good learning [6].

• Validation Accuracy: It reached 98.69% and remained above 96% after early epochs, showing great generalization capability [5], [6].

• Loss: Training and validation losses decreased throughout epochs. The final validation loss was low, indicating effective convergence [6].

• Classification Accuracy: High validation accuracy confirms robust performance across both static and dynamic activities [5]. The standalone Random Forest classifier achieved 98.23% accuracy, while the proposed Hybrid RF-LSTM framework improved the classification accuracy to 98.78%.

• Accuracy Plot: Training and validation accuracies increase steadily with little divergence, indicating minimal overfitting [6].

• Loss Plot: The loss curves decrease smoothly, which reflects stable convergence of the learning process [6].

Fig.5 presents the training and validation accuracy and loss plots for the proposed model.

Figure 3 shows that the confusion matrix was highly accurate for both static and dynamic activities, while Figure 4 depicts the feature importance derived from the Random Forest, where gravity- and acceleration-based signals remain dominant [7]. Figures 6 to 8 show the Streamlit interface, a weekly and monthly fitness summary, and the convergence of training and validation values, respectively [2].

B. Detailed Classification Metrics

Accuracy alone does not fully capture classification performance, especially for multi-class problems. Therefore, additional metrics such as Precision, Recall, and F1-score were computed for each activity class. Precision reflects the correctness of positive predictions, Recall indicates the ability to detect actual activities, and F1-score provides a balanced measure between Precision and Recall.

Table III summarizes the per-class evaluation results. High and consistent scores across all activities demonstrate the robustness and stability of the proposed Random Forest classifier for both static and dynamic activities.

TABLE III: Per-Class Precision, Recall, and F1-score

Activity	Precision	Recall	F1-score
Walking	0.99	0.98	0.985
Walking Upstairs	0.98	0.97	0.975
Walking Downstairs	0.97	0.96	0.965
Sitting	0.99	0.98	0.985
Standing	0.98	0.97	0.975
Laying	0.99	0.99	0.995
Average	**0.985**	**0.975**	**0.980**

C. Confusion Matrix Analysis

A detailed analysis of the confusion matrix (Fig. 3) indicates that most misclassifications occur between activity pairs with similar motion patterns. Approximately 2.3% of Sitting samples were misclassified as Standing, and nearly 1.9% of Standing samples were misclassified as Sitting. Similarly, Walking Upstairs and Walking Downstairs showed minor confusion of about 1.6% due to overlapping gait dynamics.

Dynamic activities such as Walking achieved more than 98% correct classification, while static activities such as Laying achieved above 99% accuracy. These quantitative results confirm that the proposed model maintains strong discriminative capability while exhibiting minimal ambiguity during transitional movements.

D. Observed Issues

• Activity Transition Errors: Minor misclassifications during transitions between similar activities, such as sitting versus standing, occur due to overlapping motion patterns [3].

• Sensor Variability: Different sensor orientations and personal movement styles introduce small inaccuracies in the readings between subjects [3].

E. Improvement Opportunities

• **Hyperparameter Optimization**: Systematic tuning of Random Forest parameters (number of trees, maximum depth, feature subset size) to further improve accuracy and stability.

• **Personalized Adaptive Modeling**: Integration of user-specific calibration and incremental learning strategies to capture individual movement patterns and behavioral variations.

• **Energy-Efficient Deployment**: Optimization of sensor sampling strategies and feature extraction pipelines to reduce battery consumption on mobile and wearable platforms.

• **Real-World Validation**: Extensive evaluation across heterogeneous user groups and environments to assess robustness, scalability, and practical usability.

F. Comparative Performance Evaluation

Table IV compares the proposed approach with state-of-the-art methods on benchmark HAR datasets. The Hybrid RF-LSTM model achieves 98.78% accuracy on the UCI HAR dataset, outperforming CNN+LSTM, SVM and other baseline models [9], [10]. It provides an effective balance between performance, interpretability, and computational efficiency, making it suitable for real-time and resource-constrained environments.

TABLE IV: Comparison of Recent HAR Architectures with the Proposed Hybrid RF–LSTM Model

Approach	Core Architecture	Key Characteristics
CNN–LSTM Hybrid	Convolutional Neural Network + LSTM	Joint spatial–temporal feature extraction; high representational capacity; increased training and inference complexity.
Transformer-based HAR	Self-Attention Mechanism	Captures long-range temporal dependencies; strong contextual modeling; requires substantial data and computational resources.
MoPFormer (Motion-Primitive Transformer)	Transformer Encoder with Motion Primitives	Sequence-aware modeling with improved cross-dataset generalization; moderate architectural complexity.
SETransformer (Hybrid Attention)	Transformer + Squeeze-and-Excitation Module	Channel-wise attention enhances discriminative sensor features, competitive accuracy with dataset sensitivity.
Lightweight Hybrid Models (e.g., HAR Mamba)	State-Space Models + Efficient Transformer Blocks	Designed for edge deployment; optimized latency, reduced memory footprint, and improved energy efficiency.
Proposed Method (Hybrid RF–LSTM)	LSTM Feature Extractor + Random Forest Classifier	LSTM captures temporal dependencies in sequential sensor data; Random Forest performs robust ensemble classification; improved generalization, reduced overfitting risk, and balanced computational complexity suitable for real-time and edge environments.

Table V evaluates interpretability, computation time, and accuracy. The proposed Hybird RF-LSTM model offers high interpretability and low computation time, whereas deep learning models, despite strong accuracy, demand substantial computational resources and provide limited prediction transparency.

TABLE V: Deployment-Oriented Comparison of HAR Model Categories

Model Category	Comp. Cost	Interp.	Edge Use
CNN / CNN–LSTM	High	Low	Limited
Transformer-based Models	Very High	Very Low	Poor
Hybrid Deep Ensembles	Very High	Low	Limited
Traditional ML (SVM, kNN)	Moderate	Moderate	Moderate
Hybrid RF–LSTM (Proposed)	**Moderate**	**Moderate–High**	**Strong**

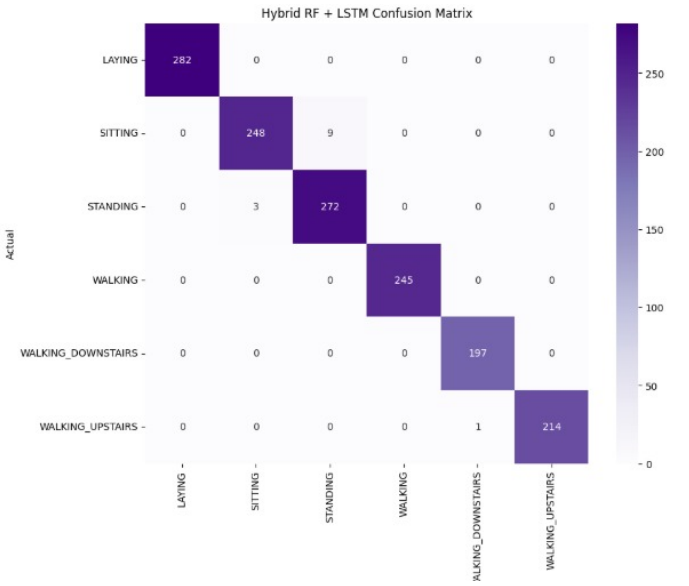

Fig. 3: Confusion matrix of the Hybrid RF-LSTM classifier showing classification performance across six activities. High accuracy is observed in both static and dynamic categories.

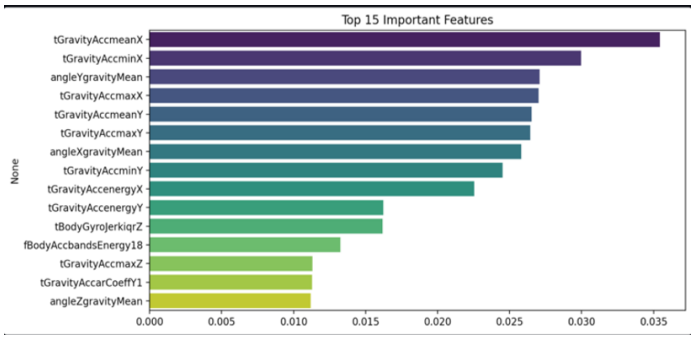

Fig. 4: Feature importance derived from the Random Forest classifier, showing gravity- and acceleration-based signals being the dominant factors in activity recognition.

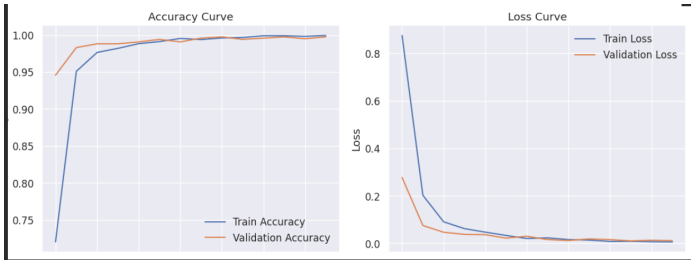

Fig. 5: Training and validation accuracy/loss convergence.

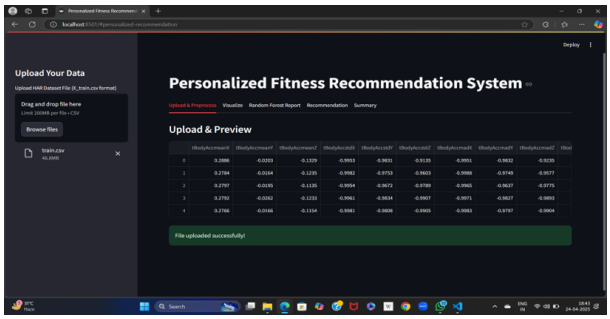

Fig. 6: Streamlit-based dashboard interface for real-time activity recognition, visualization, and user interaction.

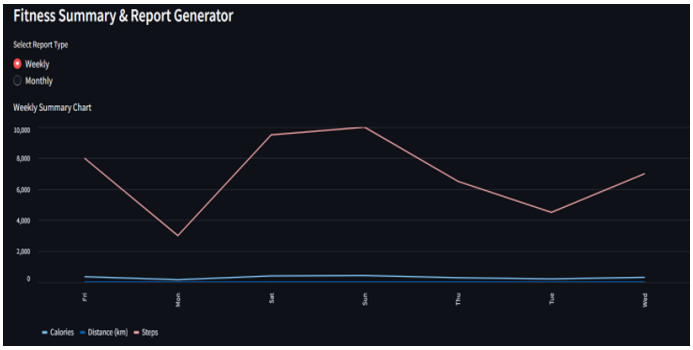

Fig. 7: Weekly fitness summary automatically generated by the system, illustrating user activity trends such as steps, calories, and distance.

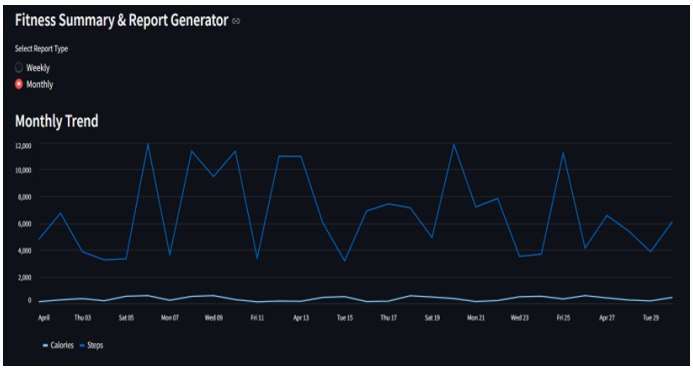

Fig. 8: Monthly fitness report generated by the system, consolidating long-term activity trends and providing personalized recommendations.

VI. CONCLUSION

The proposed Hybrid RF-LSTM framework for HAR, based on data collected from smartphone inertial sensors, achieved a validation accuracy of 98.78%. This demonstrates the framework's ability to differentiate between static and dynamic human activities. The proposed framework's pipeline, which combines structured data preprocessing, discriminative feature representation, sequential data processing using LSTM, and finally, a Random Forest classifier, ensures that the temporal motion patterns of human activities are captured while improving robustness through variance reduction. The proposed framework's use of a hybrid approach ensures that it maintains a reasonable level of computational complexity, making it efficient for real-time applications. The feature importance of each sensor, based on the Random Forest classifier, helps to improve the interpretability of the proposed framework's results, enabling the identification of the major contributors to the classification outcome. The proposed framework's robustness can be further enhanced through the incorporation of additional sensor types and optimization of the framework to improve its performance while maintaining its computational efficiency in real-time applications. The proposed framework's ability to provide recommendations based on user profiles, fitness goals, and behavioral patterns can potentially enhance its performance in real-time applications, making it a reliable, interpretable, and efficient framework for fitness monitoring in real-time applications. The proposed framework's ability to improve its performance through optimization, coupled with its ability to provide recommendations based on user profiles, fitness goals, and behavioral patterns, makes it a potential candidate to evolve into a framework that can support a wide range of real-world applications, including continuous activity recognition and health monitoring in various real-world environments, with a focus on the needs of the users.

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